CALEB ENG

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https://calebeng.github.io/

WORK EXPERIENCE

PureFacts (Summer 2024)
Software Engineering Intern

- Developed and released an Excel data-export backend module within the sprint, boosting client workflow efficiency and eliminating time-consuming manual data handling
- Created a data dictionary comprised of 100+ descriptions to facilitate client data migration, improving clarity around field mappings and enabling a faster transition to the new software
- Presented reports in daily stand up meetings to discuss progress and challenges
- Updated legacy UI graphics to align with modern design standards, enhancing visual consistency and improving perceived product quality
- Removed deprecated features and redundant UI elements, reducing user confusion and streamlining the software experience

PROJECTS

Web Development
E-Commerce Website

- Developed full stack functionality for the review, home, and store pages, ensuring consistent behavior and smooth interaction across core application features
- Designed the relational database schema to reduce duplicate data, prevent inconsistencies, and simplify future feature development
- Spearheaded the design of the SPA architecture and routing system, giving the team a unified framework that supported a consistent and cohesive user experience across all pages

Computer Vision
Emotion Detection Program

- Developed a CNN trained on facial emotion data, achieving an accuracy of 62.71%
- Co-led a group of 5, overseeing the development and testing of a CNN model variant
- Conducted background research to justify the use of a CNN over alternative ML models, selecting it for its strong performance on image based tasks and using these insights to guide dataset selection and training strategy
- Facilitated weekly team communications to monitor training progress, catch issues early, and keep the project on track with deadlines

Computer Graphics
3D graphics project

- Implemented 3d rendering using OpenGL, including camera controls, lighting, and textured object rendering
- Utilized vector normalization and spherical to cartesian conversion to implement precise camera controls and accurate projectile targeting mechanics, resulting in more consistent directional behavior

EDUCATION

2021-2025
TORONTO METROPOLITAIN UNIVERSITY

- BSc (Hons) Computer Science